

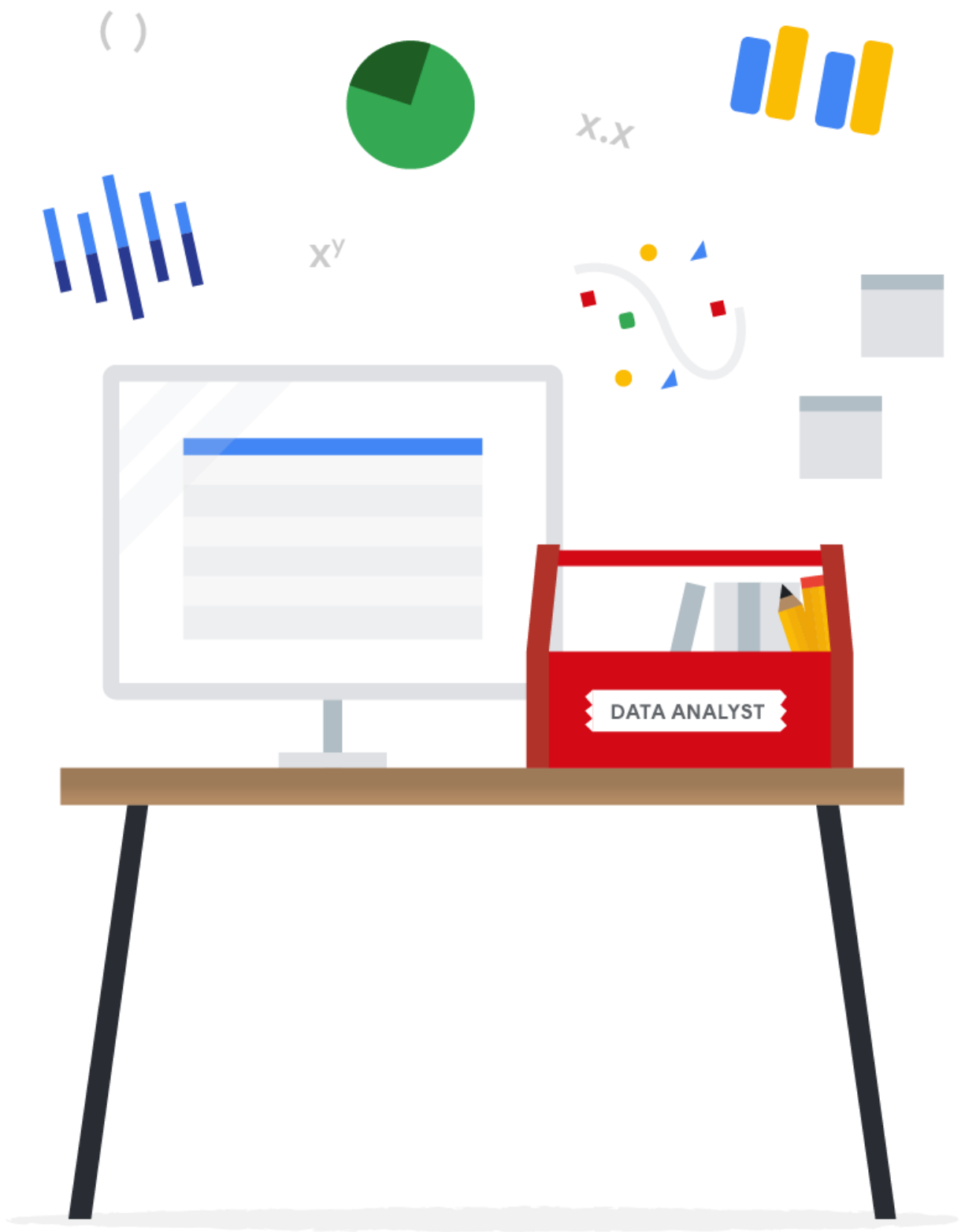


[Your Google Data Analytics Certificate roadmap](#)

Use this guide to review the topics covered, tools used, and skills you will use in each course.



Course1: Foundation



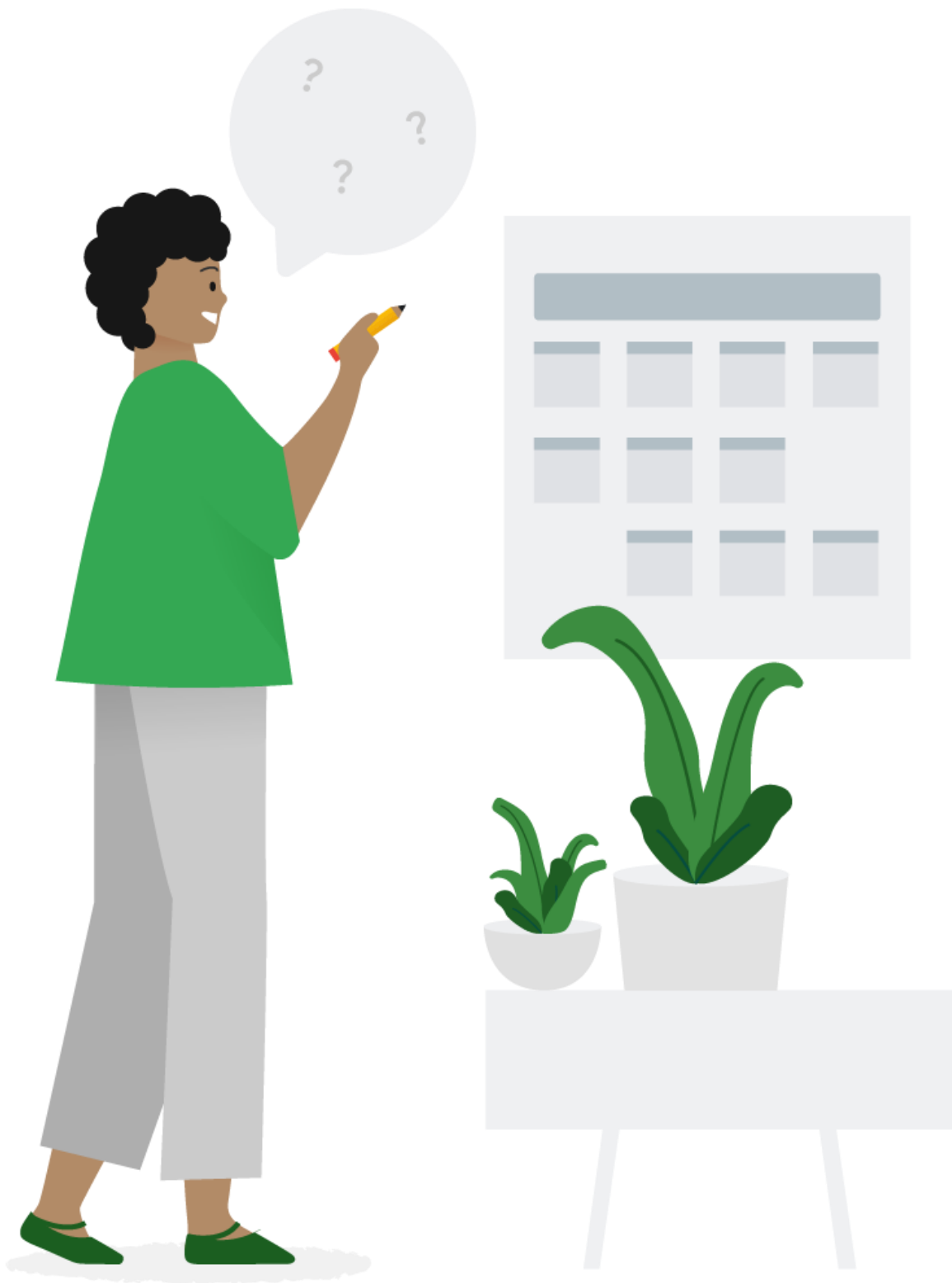
What you will learn:

- Real-life roles and responsibilities of a junior data analyst
- How businesses transform data into actionable insights
- Spreadsheet basics
- Database and query basics
- Data visualization basics

Skill sets you will build:

- Using data in everyday life
- Thinking analytically
- Applying tools from the data analytics toolkit
- Showing trends and patterns with data visualizations
- Ensuring your data analysis is fair

Course2: Ask



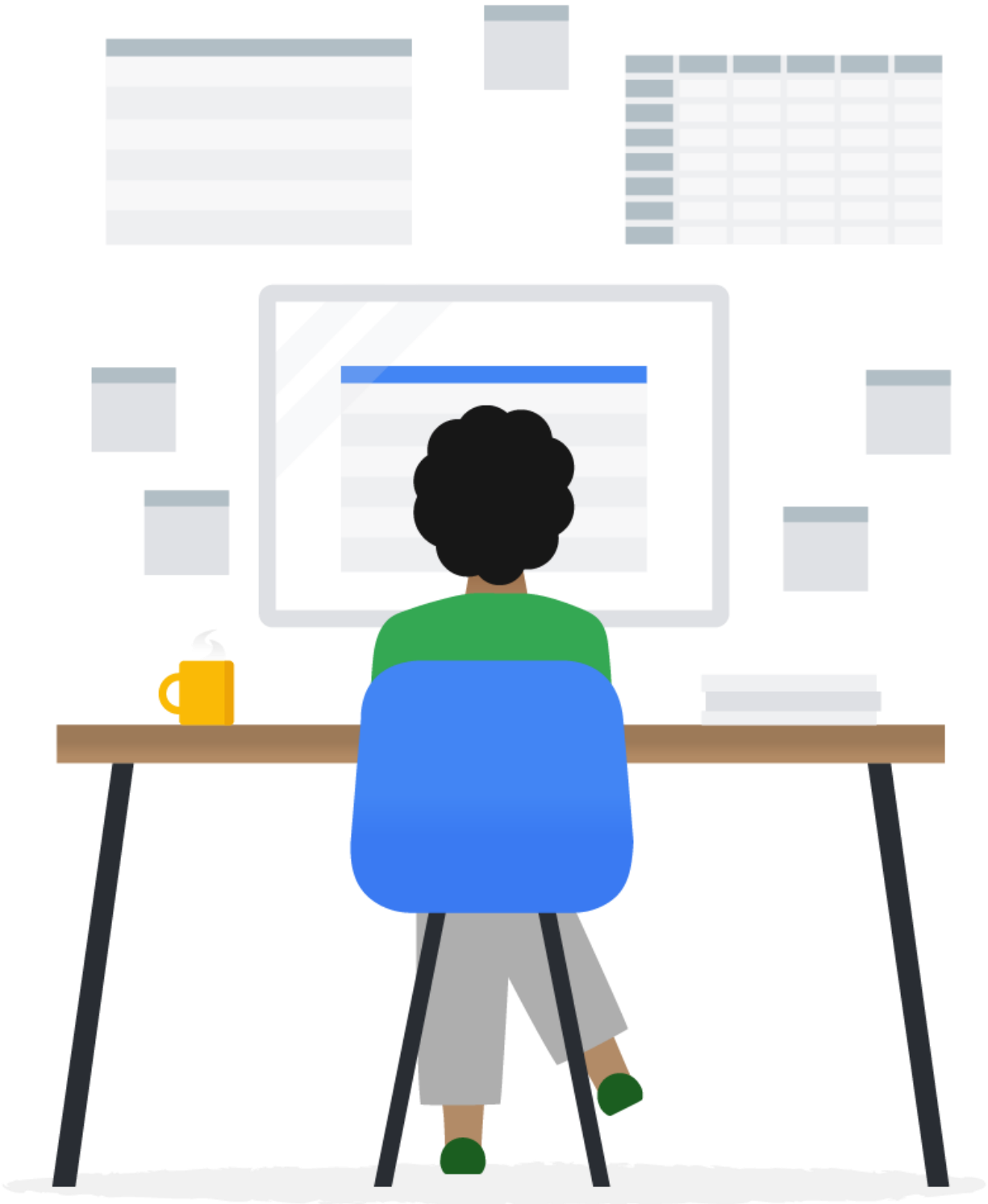
What you will learn:

- How data analysts solve problems with data
- The use of analytics for making data-driven decisions
- Spreadsheet formulas and functions
- Dashboard basics, including an introduction to Tableau
- Data reporting basics

Skill sets you will build:

- Asking SMART and effective questions
- Structuring how you think
- Summarizing data
- Putting things into context
- Managing team and stakeholder expectations
- Problem-solving and conflict-resolution

Course3: Prepare



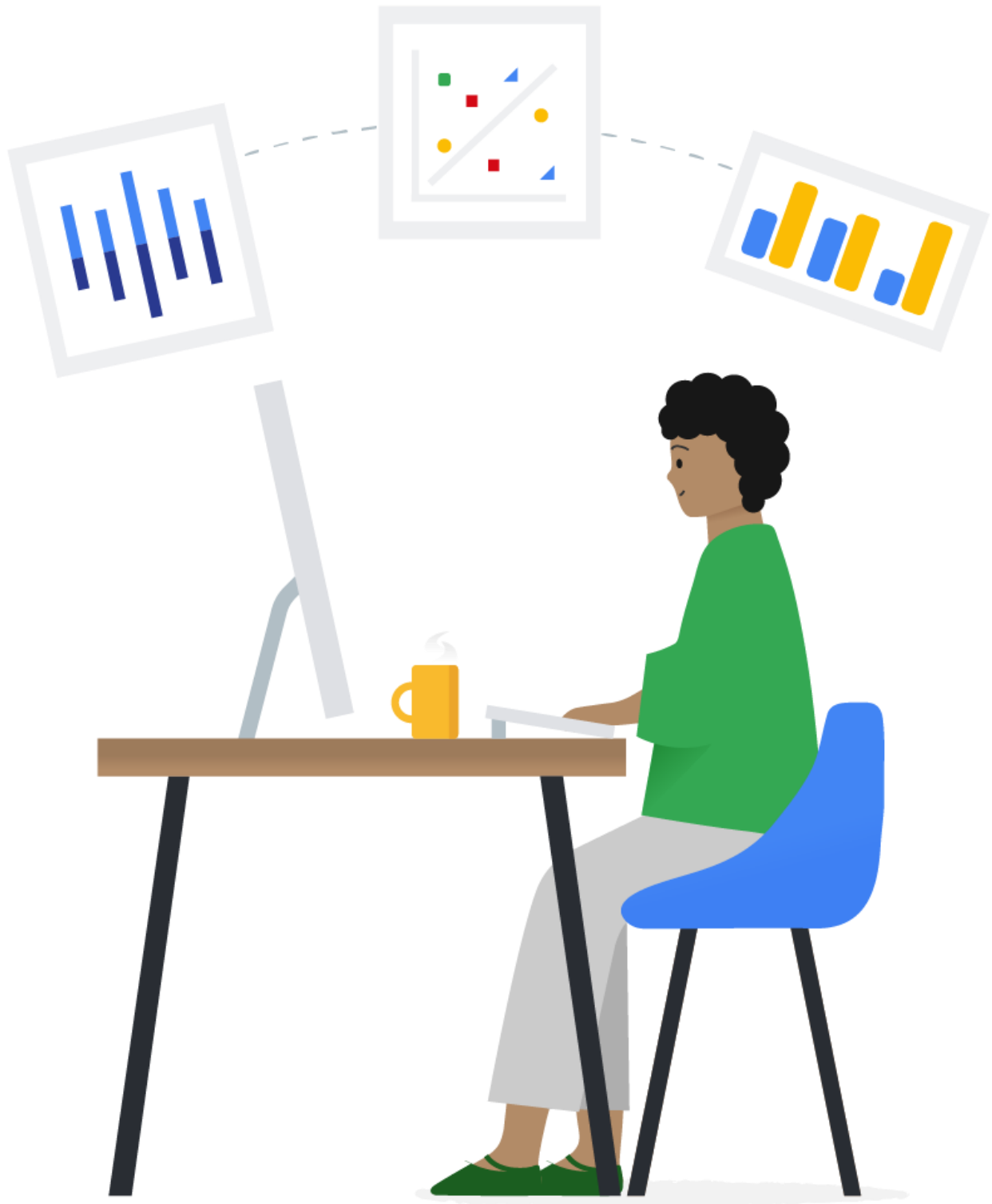
What you will learn:

- How data is generated
- Features of different data types, fields, and values
- Database structures
- The function of metadata in data analytics
- Structured Query Language (SQL) functions

Skill sets you will build:

- Ensuring ethical data analysis practices
- Addressing issues of bias and credibility
- Accessing databases and importing data
- Writing simple queries
- Organizing and protecting data
- Connecting with the data community (optional)

Course4: Process



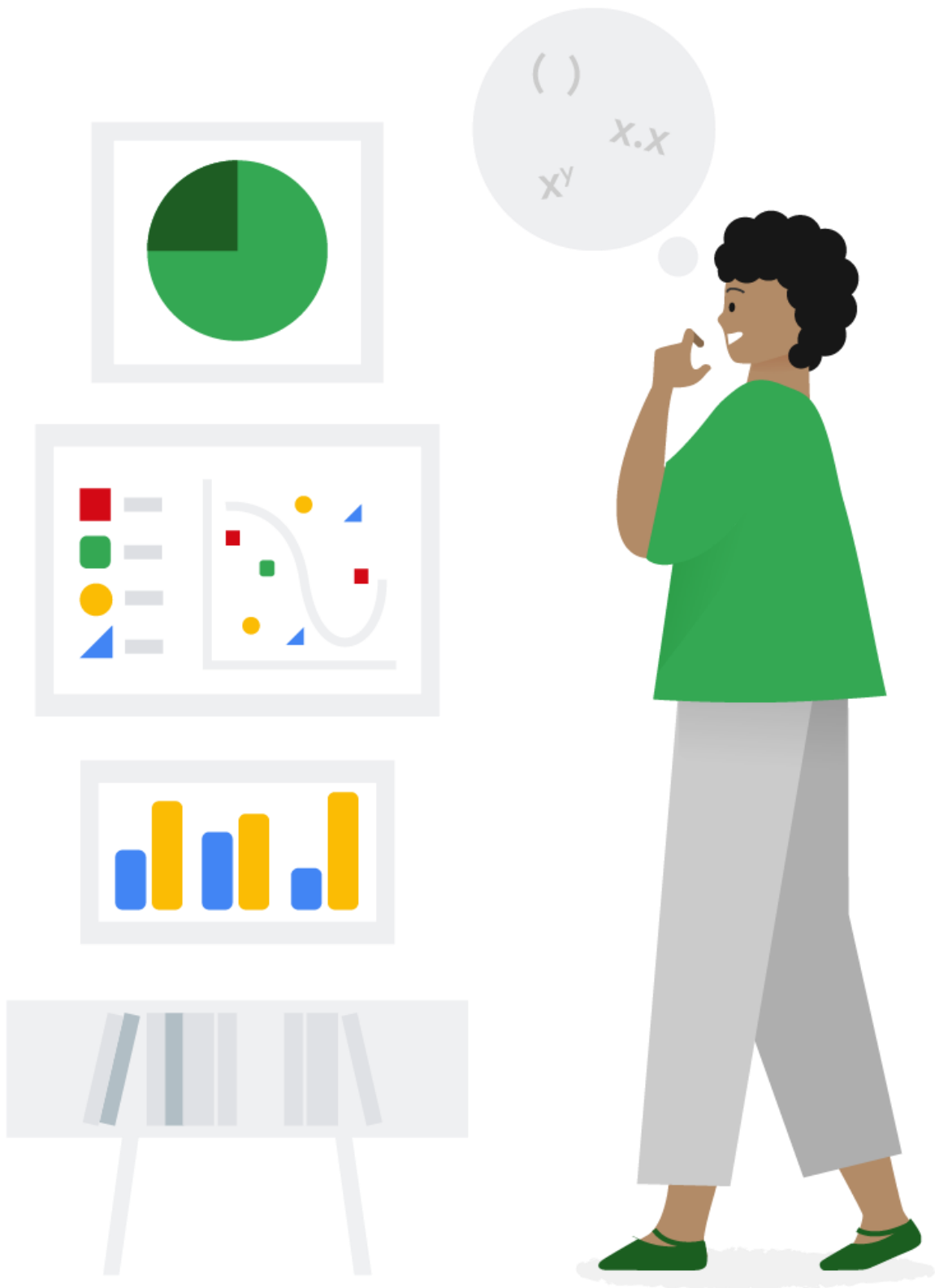
What you will learn:

- Data integrity and the importance of clean data
- The tools and processes used by data analysts to clean data
- Data-cleaning verification and reports
- Statistics, hypothesis testing, and margin of error
- Resume building and interpretation of job postings (optional)

Skill sets you will build:

- Connecting business objectives to data analysis
- Identifying clean and dirty data
- Cleaning small datasets using spreadsheet tools
- Cleaning large datasets by writing SQL queries
- Documenting data-cleaning processes

Course5: Analyze



What you will learn:

- Steps data analysts take to organize data
- How to combine data from multiple sources
- Spreadsheet calculations and pivot tables
- SQL calculations
- Temporary tables
- Data validation

Skill sets you will build:

- Sorting data in spreadsheets and by writing SQL queries
- Filtering data in spreadsheets and by writing SQL queries
- Converting data
- Formatting data
- Substantiating data analysis processes
- Seeking feedback and support from others during data analysis

Course6: Share



What you will learn:

- Design thinking
- How data analysts use visualizations to communicate about data
- The benefits of Tableau for presenting data analysis findings
- Data-driven storytelling
- Dashboards and dashboard filters
- Strategies for creating an effective data presentation

Skill sets you will build:

- Creating visualizations and dashboards in Tableau
- Addressing accessibility issues when communicating about data
- Understanding the purpose of different business communication tools
- Telling a data-driven story
- Presenting to others about data
- Answering questions about data

Course7: Act



What you will learn:

- Programming languages and environments
- R packages
- R functions, variables, data types, pipes, and vectors
- R data frames
- Bias and credibility in R
- R visualization tools
- R Markdown for documentation, creating structure, and emphasis

Skill sets you will build:

- Coding in R
- Writing functions in R
- Accessing data in R
- Cleaning data in R
- Generating data visualizations in R
- Reporting on data analysis to stakeholders

Course8: Capstone



What you will learn:

- How a data analytics portfolio distinguishes you from other candidates
- Practical, real-world problem-solving
- Strategies for extracting insights from data
- Clear presentation of data findings
- Motivation and ability to take initiative

Skill sets you will build:

- Building a portfolio
- Increasing your employability
- Showcasing your data analytics knowledge, skill, and technical expertise
- Sharing your work during an interview
- Communicating your unique value proposition to a potential employer

You're done!

You've reviewed the tools and skill sets for each course in Google Data Analytics Certificate. Feel free to return to this activity for reference.

